

HSC Practice Questions

Formulae and Equations

What is Sophie's stopping distance, to the nearest metre?

[stopping distance = reaction-time distance + braking distance]

(3 marks)

(Q28e, 2018 HSC) **Hard**

- 11 It is given that $I = \frac{3}{2}MR^2$.

What is the value of I when $M = 26.55$ and $R = 3.07$, correct to two decimal places?

- A 375.35 B 3246.08
C 9965.45 D 14 948.18 (1 mark)

(Q7, 2017 HSC) **Easy**

- 12 What is the value of x in the

equation $\frac{5-x}{3} = 6$?

- A -13 B -3
C 3 D 13 (1 mark)

(Q9, 2017 HSC) **Easy**

- 13 Young's formula, shown below, is used to calculate the dosage of medication for children aged 1–12 years based on the adult dosage.

$$D = \frac{yA}{y+12}$$

where D = dosage for children aged 1–12 years

y = age of child (in years)

A = adult dosage

A child's dosage is calculated to be 20 mg, based on an adult dosage of 40 mg.

How old is the child in years?

- A 6 B 8
C 10 D 12 (1 mark)

(Q19, 2017 HSC) **Medium**

- 14 Rhys is drinking low alcohol beer at a party over a five-hour period. He reads on the label of the low alcohol beer bottle that it is equivalent to 0.8 of a standard drink.

Rhys weighs 90 kg.

What is the maximum number of complete bottles of the low alcohol beer he can drink to remain under a Blood Alcohol Content (BAC) of 0.05?

$$[BAC]_{\text{male}} = \frac{10N - 7.5H}{6.8M}$$

where N is number of standard drinks consumed
 H is number of hours of drinking
 M is person's mass in kilograms] (4 marks)

(Q27e, 2017 HSC) **Medium**

- 15 Make y the subject of the equation

$$x = \sqrt{yp - 1}.$$

(2 marks)

(Q28d, 2017 HSC) **Medium**

- 16 Which of the following equations has $x = 5$ as the solution?

- A $x - 5 = 10$ B $5 - x = 10$
C $\frac{x}{2} = 10$ D $2x = 10$ (1 mark)

(Q2, 2016 HSC) **Easy**

- 17 Caroline drinks two small bottles of wine over a three-hour period. Each of these bottles contains 2.3 standard drinks. Caroline weighs 53 kg.

What is her approximate blood alcohol content (BAC) at the end of this period?

- A 0.081 B 0.065
C 0.0017 D 0.0014

$$[BAC]_{\text{female}} = \frac{10N - 7.5H}{5.5M}$$

where N is number of standard drinks consumed
 H is number of hours of drinking
 M is person's mass in kilograms] (1 mark)

(Q10, 2016 HSC) **Medium**

- 18 Which of the following correctly expresses

Q as the subject of $e = iR + \frac{Q}{C}$?

- A $Q = Ce + CiR$ B $Q = Ce - CiR$
C $Q = \frac{e + iR}{C}$ D $Q = \frac{e - iR}{C}$ (1 mark)

(Q24, 2016 HSC) **Medium**

- 19 The number of 'standard drinks' in various glasses of wine is shown.

Number of standard drinks

White Wine		Red Wine	
small glass	large glass	small glass	large glass
0.9	1.4	1.0	1.5

A woman weighing 62 kg drinks three small glasses of white wine and two large glasses of red wine between 8 pm and 1 am.

What would be her blood alcohol content (BAC) estimate at 1 am, correct to three decimal places?

- A 0.030 B 0.037
C 0.046 D 0.057

$$[BAC]_{\text{female}} = \frac{10N - 7.5H}{5.5M}$$

where N is number of standard drinks consumed
 H is number of hours of drinking
 M is person's mass in kilograms] (1 mark)

(Q23, 2015 HSC) **Medium**

20 Consider the equation $\frac{2x}{3} - 4 = \frac{5x}{2} + 1$.

Which of the following would be a correct step in solving this equation?

- A $\frac{2x}{3} - 3 = \frac{5x}{2}$ B $\frac{2x}{3} = \frac{5x}{2} + 5$
 C $2x - 4 = \frac{15x}{2} + 3$ D $\frac{4x}{6} - 8 = 5x + 2$ (1 mark)

(Q24, 2015 HSC) Medium

- 21 Clark's formula is used to determine the dosage of medicine for children.

$$\text{Dosage} = \frac{\text{weight in kg} \times \text{adult dosage}}{70}$$

The adult daily dosage of a medicine contains 3150 mg of a particular drug.

A child who weighs 35 kg is to be given tablets each containing 525 mg of this drug.

How many tablets should this child be given daily? (2 marks)

(Q26b, 2015 HSC) Easy

- 22 The formula $C = \frac{5}{9}(F - 32)$ is used to convert temperatures between degrees Fahrenheit (F) and degrees Celsius (C).

Convert 3°C to the equivalent temperature in Fahrenheit. (2 marks)

(Q28d, 2015 HSC) Easy

- 23 Claire is driving on a motorway at a speed of 110 kilometres per hour and has to brake suddenly. She has a reaction time of 2 seconds and a braking distance of 59.2 metres.

Calculate her stopping distance.

[stopping distance = reaction-time distance + braking distance] (2 marks)

(Q30d, 2015 HSC) Hard

- 24 Young's formula below is used to calculate the required dosages of medicine for children aged 1–12 years.

$$\text{Dosage} = \frac{\text{age of child (in years)} \times \text{adult dosage}}{\text{age of child (in years)} + 12}$$

How much of the medicine should be given to an 18-month-old child in a 24-hour period if each adult dosage is 45 mL? The medicine is to be taken every 6 hours by both adults and children.

- A 5 mL B 20 mL
 C 27 mL D 30 mL (1 mark)

(Q4, 2014 HSC) Medium

25 Solve the equation $\frac{5x+1}{3} - 4 = 5 - 7x$. (3 marks)

(Q26c, 2014 HSC) Medium

- 26 What is the maximum number of standard drinks that a male weighing 84 kg can consume over 4 hours in order to maintain a blood alcohol content (BAC) of less than 0.05?

$$[\text{BAC}]_{\text{male}} = \frac{10N - 7.5H}{6.8M}$$

where N is number of standard drinks consumed
 H is number of hours of drinking
 M is person's mass in kilograms] (3 marks)

(Q29b, 2014 HSC) Medium

- 27 Which equation correctly shows r as the subject of $S = 800(1 - r)$?

- A $r = \frac{800 - S}{800}$ B $r = \frac{S - 800}{800}$
 C $r = 800 - S$ D $r = S - 800$ (1 mark)

(Q21, 2013 HSC) Hard

- 28 Sarah tried to solve this equation and made a mistake in Line 2.

$$\frac{W+4}{3} - \frac{2W-1}{5} = 1 \quad \dots\dots \text{Line 1}$$

$$5W + 20 - 6W - 3 = 15 \quad \dots\dots \text{Line 2}$$

$$17 - W = 15 \quad \dots\dots \text{Line 3}$$

$$W = 2 \quad \dots\dots \text{Line 4}$$

Copy the equation in Line 1 into your writing booklet and continue your solution to solve this equation for W . Show all lines of working. (2 marks)

(Q29a, 2013 HSC) Hard

- 29 Which of the following correctly expresses c as the subject of $E = mc^2 + p$?

- A $c = \pm \sqrt{\frac{E-p}{m}}$ B $c = \pm \sqrt{\frac{E+p}{m}}$
 C $c = \pm \sqrt{\frac{E}{m}} - p$ D $c = \pm \sqrt{\frac{E}{m} - p}$ (1 mark)

(Q21, 2012 HSC) Medium

- 30 Which of the following correctly expresses a as the subject of $s = ut + \frac{1}{2}at^2$?

- A $a = \frac{2(s-ut)}{t^2}$ B $a = \frac{2s-ut}{t^2}$
 C $a = \frac{\frac{1}{2}(s-ut)}{t^2}$ D $a = \frac{\frac{1}{2}s-ut}{t^2}$ (1 mark)

(Q18, 2011 HSC) Hard

HSC Practice Questions

Formulae and Equations

- 31 If $M = -9$, what is the value of $\frac{3M^2 + 5M}{6}$?
- A -250.5 B -48
C 33 D 235.5 (1 mark)
- (Q7, 2010 HSC) Medium

- 32 Which of the following correctly expresses x as the subject of $a = \frac{nx}{5}$?
- A $x = \frac{an}{5}$ B $x = \frac{5a}{n}$
C $x = \frac{a-5}{n}$ D $x = 5a - n$ (1 mark)
- (Q18, 2010 HSC) Medium

- 33 Fred tried to solve this equation and made a mistake in Line 2.

$$\begin{array}{ll} 4(y+2) - 3(y+1) = -3 & \text{Line 1} \\ 4y + 8 - 3y + 3 = -3 & \text{Line 2} \\ y + 11 = -3 & \text{Line 3} \\ y = -14 & \text{Line 4} \end{array}$$

Copy the equation in Line 1 into your writing booklet.

- i Rewrite Line 2 correcting his mistake. (1 mark) Easy
- ii Continue your solution showing the correct working for Lines 3 and 4 to solve this equation for y . (1 mark) Easy

(Q24a, 2010 HSC)

- 34 Which of the following correctly expresses n as the subject of $v = \frac{3mn^2}{r}$?
- A $n = \pm \frac{\sqrt{rv}}{3m}$ B $n = \pm r\sqrt{\frac{v}{3m}}$
C $n = \pm \frac{r\sqrt{v}}{3m}$ D $n = \pm \sqrt{\frac{rv}{3m}}$ (1 mark)
- (Q15, 2009 HSC) Medium

- 35 What is the value of $\sqrt{\frac{x+2y}{8y}}$ if $x = 5.6$ and $y = 3.1$, correct to 2 decimal places?
- A 0.69 B 2.62
C 2.83 D 4.77 (1 mark)
- (Q9, 2008 HSC) Easy

- 36 Solve $\frac{5x+1}{2} = 4x-7$. (3 marks)
- (Q23d, 2008 HSC) Medium

- 37 Which of the following correctly expresses T as the subject of $B = 2\pi\left(R + \frac{T}{2}\right)$?

- A $T = \frac{B}{\pi} - 2R$ B $T = \frac{B}{\pi} - R$
C $T = 2R - \frac{B}{\pi}$ D $T = \frac{B}{4\pi} - \frac{R}{2}$ (1 mark)
- (Q19, 2007 HSC) Hard

- 38 The distance in kilometres (D) of an observer from the centre of a thunderstorm can be estimated by counting the number of seconds (t) between seeing the lightning and first hearing the thunder.

Use the formula $D = \frac{t}{3}$ to estimate the number of seconds between seeing the lightning and hearing the thunder if the storm is 1.2 km away. (1 mark)

(Q24b, 2007 HSC) Easy

- 39 If $V = \frac{4}{3}\pi r^3$, what is the value of V when $r = 2$, correct to two decimal places?
- A 8.38 B 12.57
C 25.13 D 33.51 (1 mark)
- (Q2, 2006 HSC) Easy

- 40 What is the formula for q as the subject of $4p = 5t + 2q^2$?
- A $q = \pm \frac{\sqrt{5t-4p}}{2}$ B $q = \pm \frac{\sqrt{4p-5t}}{2}$
C $q = \pm \sqrt{\frac{5t-4p}{2}}$ D $q = \pm \sqrt{\frac{4p-5t}{2}}$ (1 mark)
- (Q18, 2006 HSC) Hard

- 41 What is the value of $\frac{a-b}{4}$, if $a = 240$ and $b = 56$?
- A 4 B 46
C 226 D 736 (1 mark)
- (Q2, 2005 HSC) Easy

- 42 Using the formula $d = 5t^3 - 2$, Marcia tried to find the value of t when $d = 137$.

Here is her solution. She has made one mistake.

$d = 5t^3 - 2$		Line A
$137 = 5t^3 - 2$		Line B
$135 = 5t^3$		Line C
$27 = t^3$		Line D
$t = 3$		

Which line does NOT follow correctly from the previous line?

HSC Practice Questions

Formulae and Equations

- A Line A
C Line C

- B Line B
D Line D (1 mark)

(Q14, 2005 HSC) Easy

- 43 The formula $D = \frac{2A}{15}$ is used to calculate the dosage of Hackalot cough medicine to be given to a child.
- D is the dosage of Hackalot cough medicine in millilitres (mL).
 - A is the age of the child in months.
- i If George is nine months old, what dosage of Hackalot cough medicine should he be given? (1 mark) Easy
- ii The correct dosage of Hackalot cough medicine for Sam is 4 mL. What is the difference in the ages of Sam and George, in months? (3 marks) Hard

(Q24b, 2005 HSC)

- 44 Make L the subject of the equation $T = 2\pi L^2$. (2 marks) Medium
- (Q24c, 2005 HSC)

- 45 If $K = Ft^3$, $F = 5$ and $t = 0.715$, what is the value of K correct to three significant figures?
- A 1.82 B 1.827
C 1.828 D 1.83 (1 mark)
- (Q3, 2004 HSC) Easy

- 46 If $d = 6t^2$, what is a possible value of t when $d = 2400$?
- A 0.05 B 20
C 120 D 400 (1 mark)
- (Q11, 2004 HSC) Easy

- 47 Kirbee is shopping for computer software. *Novirus* costs \$115 more than *Funmaths*. Let x dollars be the cost of *Funmaths*.
- i Write an expression involving x for the cost of *Novirus*. (1 mark) Easy
- ii *Novirus* and *Funmaths* together cost \$415. Write an equation involving x and solve it to find the cost of *Funmaths*. (2 marks) Medium
- (Q23b, 2004 HSC)

- 48 A health rating, R , is calculated by dividing a person's weight, w , in kilograms by the square of the person's height, h , in metres.
- i Fred is 150 cm and weighs 72 kg. Calculate Fred's health rating. (1 mark) Easy
- ii Over several years, Fred expects to grow 10 cm taller. By this time he wants his health rating to be 25. How much weight should he gain or lose to achieve his aim?

Justify your answer with mathematical calculations. (2 marks) Medium

(Q28a, 2004 HSC)

- 49 If $d = \sqrt{\frac{h}{5}}$, what is the value of d , correct to one decimal place, when $h = 28$?
- A 1.1 B 2.4
C 2.8 D 5.6 (1 mark)
- (Q4, 2003 HSC) Easy

- 50 Kylie and Danny work in a music store. The weekly wage \$ W of an employee at the store is given by

$$W = 0.75n + 50,$$

where n is the number of CDs the employee sells.

If Kylie sells two more CDs than Danny in one week, how much more will she earn?

- A \$0.75 B \$1.50
C \$50.75 D \$51.50 (1 mark)
- (Q15, 2003 HSC) Easy

- 51 i While Sandra is on holiday she visits countries where the Fahrenheit temperature scale is used. She knows that the correct way to convert from Celsius to Fahrenheit is:

'Multiply the Celsius temperature by 1.8, then add 32.'

Find the value of A in the following table.

Celsius	Fahrenheit
5	41
15	59
A	77

(1 mark) Medium

- ii Peter uses the following method to approximate the conversion from Celsius to Fahrenheit:

'Add 12 to the Celsius temperature, then double your result.'

Express Peter's rule as an algebraic equation. Use C for the Celsius temperature and F for the approximate Fahrenheit temperature. (2 marks) Medium

(Q28c, 2003 HSC)



Prelim

HSC Practice Questions

Formulae and Equations

52 If $w = 2y^3 - 1$, what is the value of y when $w = 13$?

A $\frac{\sqrt[3]{14}}{2}$

B $\sqrt[3]{6}$

C $\sqrt[3]{7}$

D $\sqrt[3]{14}$ (1 mark)

(Q16, 2002 HSC) Easy

53 If $w = \frac{15y}{y + 12}$, and $y = 7$, find the value of w (correct to two decimal places).

A 5.53

B 8.26

C 15.75

D 27.00 (1 mark)

(Q2, 2001 HSC) Easy

1 $a = \frac{b}{7}$
When $b = 22$,
 $a = \frac{22}{7}$
 $= 3.14285714...$
 $= 3.14$ (3 significant figures)
Answer A

2 The number is x .
Subtracting 8 gives $x - 8$.
Then multiplying by 3 gives $3(x - 8)$.
2 more than x is $x + 2$.
So the equation is $3(x - 8) = x + 2$.
Answer C

3 $x + \frac{x-1}{2} = 9$
Multiplying by 2:
 $2x + x - 1 = 18$ ✓
 $3x - 1 = 18$
 $3x = 19$
 $x = 6\frac{1}{3}$ ✓ (2 marks)

4 Time = $\frac{BAC}{0.015}$
If $BAC = 0.08375$
Time = $\frac{0.08375}{0.015}$
 $= 5 \text{ h } 35 \text{ min}$
So, according to the formula,
Jake's BAC will be zero
5 h 35 min after 10:30 pm or at
4:05 am.
If $BAC = 0.05$,
Time = $\frac{0.05}{0.015}$
 $= 3 \text{ h } 20 \text{ min}$
According to the formula, BAC
will reach 0 from 0.05 after 3 h
20 min.
So Jake might assume that his
BAC is 0.05 3 h 20 min before
4:05 am, so at 12:45 am.
Answer A

5 $3x - 4y - 1 = 0$
 $4y = 3x - 1$
 $y = \frac{3x-1}{4}$
Answer C

6 $M = 60$
 $N = 3 \times 1.2 + 4$
 $= 76$ ✓
From 6:15 pm until 12:30 am is
 $6\frac{1}{4}$ hours.

So $H = 6.25$
 $BAC = \frac{10N - 7.5H}{5.5M}$
 $= \frac{10 \times 7.6 - 7.5 \times 6.25}{5.5 \times 60}$
 $= 0.088257575...$ ✓
Time = $\frac{BAC}{0.015}$
 $= 5.88383838... \text{ h}$
 $= 5 \text{ h } 53 \text{ min}$ (nearest minute) ✓
Now 5 h 53 min after 12:30 am is
6:23 am.
So, using these formulae,
Hannah's BAC should reach zero
at 6:23 am. ✓ (4 marks)

7 $k = \frac{1}{2}mv^2$
 $2k = mv^2$
 $v^2 = \frac{2k}{m}$
 $v = \pm \sqrt{\frac{2k}{m}}$
Answer A

8 Dosage = 90 mg,
adult dosage = 325 mg
Dosage = $\frac{\text{weight in kg} \times \text{adult dosage}}{70}$
 $90 = \frac{\text{weight in kg} \times 325}{70}$ ✓
 $6300 = \text{weight in kg} \times 325$
Weight = $19.3846153... \text{ kg}$
 $= 19 \text{ kg}$ (nearest kg) ✓ (2 marks)

9 $\frac{2x+1}{5} = \frac{3x+1}{2}$
 $10 \times \frac{2x+1}{5} + 10 \times 1 = \frac{10(3x+1)}{2}$ ✓
 $4x + 10 = 15x + 5$ ✓
 $5 = 11x$
 $x = \frac{5}{11}$ (3 marks) ✓

10 $70 \text{ km/h} = 70000 \text{ m/h}$
 $= 19.4444444... \text{ m/s}$
In 1.5 s Sophie travels
 $1.5 \times 19.4444444... \text{ m}$
So reaction time distance
 $= 29.1666666... \text{ m}$
 $= 29 \text{ m}$ (nearest metre) ✓
Braking distance:
 $D = 0.01v^2$
 $= 0.01 \times 70^2$
 $= 49 \text{ m}$ ✓
Stopping distance
 $= (29 + 49) \text{ m} = 78 \text{ m}$
Sophie's stopping distance is
78 m, to the nearest
metre. ✓ (3 marks)

11 $I = \frac{3}{2}MR^2$
When $M = 26.55$ and $R = 3.07$,
 $I = \frac{3}{2} \times 26.55 \times 3.07^2$
 $= 375.346642...$
 $= 375.35$ (2 d.p.)
Answer A

12 $\frac{5-x}{3} = 6$
 $5-x = 18$
 $-x = 13$
 $x = -13$
Answer A

13 $D = \frac{yA}{y+12}$
When $A = 40$, $D = 20$
 $20 = \frac{40y}{y+12}$
 $20y + 240 = 40y$
 $240 = 20y$
 $12 = y$
Answer D

14 $BAC_{\text{male}} = \frac{10N - 7.5H}{6.8M}$
If $BAC = 0.05$, $H = 5$ and
 $M = 90$,
 $0.05 = \frac{10N - 7.5 \times 5}{6.8 \times 90}$ ✓
 $= \frac{10N - 37.5}{612}$
 $30.6 = 10N - 37.5$
 $68.1 = 10N$
 $6.81 = N$ ✓
So the number of standard
drinks must be less than 6.81.
Number of bottles = $6.81 \div 0.8$
 $= 8.5125$ ✓
So the maximum number of
complete bottles Rhys can
drink in 5 hours to stay under
0.05 is 8. ✓ (4 marks)

15 $x = \sqrt{yp-1}$
 $x^2 = yp - 1$ ✓
 $x^2 + 1 = yp$
 $\frac{x^2+1}{p} = y$
 $y = \frac{x^2+1}{p}$ ✓ (2 marks)

16 $2x = 10$
 $x = 5$
Answer D

17 $N = 2 \times 2.3 = 4.6$
 $H = 3$ $M = 53$

$$\begin{aligned} BAC_{\text{female}} &= \frac{10N - 7.5H}{5.5M} \\ &= \frac{10 \times 4.6 - 7.5 \times 3}{5.5 \times 53} \\ &= 0.0806174957... \\ &= 0.081 \quad (2 \text{ sig. figs.}) \end{aligned}$$

Answer A

$$\begin{aligned} 18 \quad e &= iR + \frac{Q}{C} \\ \frac{Q}{C} &= e - iR \\ Q &= Ce - CiR \end{aligned}$$

Answer B

$$\begin{aligned} 19 \quad \text{Number of standard drinks} &= 3 \times 0.9 + 2 \times 1.5 \\ &= 5.7 \end{aligned}$$

$$\begin{aligned} BAC_{\text{female}} &= \frac{10N - 7.5H}{5.5M} \\ &= \frac{10 \times 5.7 - 7.5 \times 5}{5.5 \times 62} \\ &= 0.05718475... \\ &= 0.057 \quad (3 \text{ d.p.}) \end{aligned}$$

Answer D

$$\begin{aligned} 20 \quad \frac{2x}{3} - 4 &= \frac{5x}{2} + 1 \\ \text{Add 4 to both sides:} \\ \frac{2x}{3} &= \frac{5x}{2} + 5 \end{aligned}$$

Answer B

$$\begin{aligned} 21 \quad \text{Dosage} &= \frac{\text{weight in kg} \times \text{adult dosage}}{70} \\ &= \frac{35 \times 3150}{70} \\ &= 1575 \text{ mg} \\ \text{Number of tablets} &= 1575 \div 525 \\ &= 3 \\ \text{The child should be given} &= 3 \text{ tablets daily.} \end{aligned}$$

(2 marks)

$$\begin{aligned} 22 \quad C &= \frac{5}{9}(F - 32) \\ 3 &= \frac{5}{9}(F - 32) \\ 5.4 &= F - 32 \\ 37.4 &= F \\ \text{So } 3^\circ\text{C is equivalent to } &37.4^\circ\text{F.} \end{aligned}$$

(2 marks)

$$\begin{aligned} 23 \quad \text{Speed} &= 110 \text{ km/h} \\ &= 110000 \text{ m/h} \\ &= 30.5555... \text{ m/s} \\ \text{Reaction time distance} &= 2 \times 30.5555... \\ &= 61.1111... \\ &= 61.1 \text{ m} \quad (1 \text{ d.p.}) \end{aligned}$$

$$\begin{aligned} \text{Stopping distance} &= (61.1 + 59.2) \text{ m} \\ &= 120.3 \text{ m} \end{aligned}$$

(2 marks)

$$\begin{aligned} 24 \quad \text{Each dose} &= \frac{1.5 \times 45}{1.5 + 12} \\ &= 5 \text{ mL} \\ \text{Number of doses} &= 24 \div 6 \\ &= 4 \\ \text{Total medicine} &= 4 \times 5 \text{ mL} \\ &= 20 \text{ mL} \end{aligned}$$

Answer B

$$\begin{aligned} 25 \quad \frac{5x + 1}{3} - 4 &= 5 - 7x \\ 5x + 1 - 12 &= 15 - 21x \\ 26x &= 26 \\ x &= 1 \end{aligned}$$

(3 marks)

$$\begin{aligned} 26 \quad BAC_{\text{male}} &= \frac{10N - 7.5H}{6.8M} \\ 0.05 &= \frac{10N - 7.5 \times 4}{6.8 \times 84} \\ &= \frac{10N - 30}{571.2} \\ 10N - 30 &= 28.56 \\ 10N &= 58.56 \\ N &= 5.856 \end{aligned}$$

So the maximum number of standard drinks to stay under 0.05 would be 5.

(3 marks)

$$\begin{aligned} 27 \quad S &= 800(1 - r) \\ \frac{S}{800} &= 1 - r \\ r &= 1 - \frac{S}{800} \\ &= \frac{800 - S}{800} \end{aligned}$$

Answer A

$$\begin{aligned} 28 \quad \frac{W + 4}{3} - \frac{2W - 1}{5} &= 1 \\ 5W + 20 - 6W + 3 &= 15 \\ 23 - W &= 15 \\ W &= 8 \end{aligned}$$

(2 marks)

$$\begin{aligned} 29 \quad E &= mc^2 + p \\ mc^2 + p &= E \\ mc^2 &= E - p \\ c^2 &= \frac{E - p}{m} \\ c &= \pm \sqrt{\frac{E - p}{m}} \end{aligned}$$

Answer A

$$\begin{aligned} 30 \quad s &= ut + \frac{1}{2}at^2 \\ s - ut &= \frac{1}{2}at^2 \\ 2(s - ut) &= at^2 \\ \frac{2(s - ut)}{t^2} &= a \\ \text{So } a &= \frac{2(s - ut)}{t^2} \end{aligned}$$

Answer A

$$\begin{aligned} 31 \quad \text{If } M &= -9, \\ \frac{3M^2 + 5M}{6} &= \frac{3 \times (-9)^2 + 5 \times (-9)}{6} \\ &= \frac{243 - 45}{6} \\ &= 33 \end{aligned}$$

Answer C

$$\begin{aligned} 32 \quad a &= \frac{nx}{5} \\ [\text{Multiply both sides by 5}] \\ 5a &= nx \\ [\text{Divide both sides by } n.] \\ \frac{5a}{n} &= x \\ \text{So } x &= \frac{5a}{n} \end{aligned}$$

Answer B

$$\begin{aligned} 33 \quad 4(y + 2) - 3(y + 1) &= -3 \\ \text{i } 4y + 8 - 3y - 3 &= -3 \quad (1 \text{ mark}) \\ \text{ii } y + 5 &= -3 \\ y &= -8 \quad (1 \text{ mark}) \end{aligned}$$

$$\begin{aligned} 34 \quad v &= \frac{3mn^2}{r} \\ rv &= 3mn^2 \\ \frac{rv}{3m} &= n^2 \\ \therefore n &= \pm \sqrt{\frac{rv}{3m}} \end{aligned}$$

Answer D

$$\begin{aligned} 35 \quad \sqrt{\frac{x + 2y}{8y}} &= \sqrt{\frac{5.6 + 2 \times 3.1}{8 \times 3.1}} \\ &= 0.6897... \quad (\text{by calc.}) \\ &\approx 0.69. \end{aligned}$$

Answer A

$$\begin{aligned} 36 \quad \frac{5x + 1}{2} &= 4x - 7 \\ 5x + 1 &= 2(4x - 7) \\ 5x + 1 &= 8x - 14 \\ 5x - 8x &= -14 - 1 \\ -3x &= -15 \\ x &= \frac{-15}{-3} \\ &= 5 \end{aligned}$$

(3 marks)

37 $B = 2\pi\left(R + \frac{T}{2}\right)$

$$\begin{aligned}\therefore \frac{B}{2\pi} &= R + \frac{T}{2} \\ \therefore \frac{B}{2\pi} - R &= \frac{T}{2} \\ \therefore T &= 2\left(\frac{B}{2\pi} - R\right) \\ &= \frac{B}{\pi} - 2R\end{aligned}$$

Answer A

38 $D = \frac{t}{3}$

$$\begin{aligned}\therefore 1.2 &= \frac{t}{3} \\ \therefore t &= 3.6 \text{ seconds}\end{aligned}$$

(1 mark)

39 $V = \frac{4}{3}\pi r^3$

$$\begin{aligned}\therefore V &= \frac{4}{3} \times \pi \times 2^3 \\ &= 33.5103 \dots \\ &= 33.51 \text{ to 2 decimal places}\end{aligned}$$

Answer D

40 $4p = 5t + 2q^2$

$$\begin{aligned}2q^2 &= 4p - 5t \\ q^2 &= \frac{4p - 5t}{2} \\ q &= \pm \sqrt{\frac{4p - 5t}{2}}\end{aligned}$$

Answer D

41 $\frac{a-b}{4} = \frac{240-56}{4}$

$$= 46$$

Answer B

42 $d = 5t^3 - 2$

$$\begin{aligned}\therefore 137 &= 5t^3 - 2 \quad \dots \text{A} \\ 139 &= 5t^3 \quad \dots \text{B}\end{aligned}$$

$\therefore$ Line B is incorrect.

Answer B

43 i $D = \frac{2A}{15}$

If $A = 9$,

$$\begin{aligned}D &= \frac{2 \times 9}{15} \\ &= 1.2 \text{ mL}\end{aligned}$$

(1 mark)

ii If $D = 4$,

$$4 = \frac{2A}{15}$$

$$4 \times 15 = 2A$$

$$60 = 2A$$

$$\therefore A = 30$$

$\therefore$ Sam is 30 months ✓

$\therefore$ Difference in the ages

$$= 30 - 9$$

$$= 21 \text{ months} \quad \checkmark$$

(3 marks)

44 $T = 2\pi L^2$

$$2\pi L^2 = T$$

$$L^2 = \frac{T}{2\pi}$$

$$\therefore L = \pm \sqrt{\frac{T}{2\pi}}$$

(2 marks)

45 $K = Fr^3$

$$= 5 \times 0.715^3$$

$$= 1.8276 \dots$$

$$= 1.83 \text{ to 3 sig. figures}$$

Answer D

46 $d = 6t^2$

$$\therefore 2400 = 6t^2$$

$$t^2 = \frac{2400}{6}$$

$$= 400$$

$$\therefore t = \pm \sqrt{400}$$

$$= \pm 20$$

Answer B

47 i Cost of Novirus

= \$115 more than x

$$= \$115 + x$$

(1 mark)

ii $x + 115 + x = 415$ ✓

$$2x = 415 - 115$$

$$= 300$$

$$\therefore x = 150$$

$\therefore$ Cost of Funmaths is \$150. ✓

(2 marks)

48 i $R = \frac{w}{h^2}$

If $h = 150 \text{ cm} = 1.5 \text{ m}$

and $w = 72 \text{ kg}$

$$\therefore R = \frac{72}{1.5^2}$$

$$= 32$$

(1 mark)

ii If $R = 25$

and $h = 1.6 \text{ m}$

$$\therefore 25 = \frac{w}{1.6^2}$$

$$w = 25 \times 1.6^2$$

$$= 64 \text{ kg}$$

$\therefore$ Fred should lose 8 kg. ✓

(2 marks)

49 $d = \sqrt{\frac{h}{5}}$

If $h = 28$,

$$d = \sqrt{\frac{28}{5}}$$

$$= \sqrt{5.6}$$

$$= 2.366 \dots$$

$$= 2.4 \text{ to 1 decimal place}$$

Answer B

50 If Danny sells 1 CD,

$$W = \$0.75 \times 1 + \$50$$

$$= \$50.75$$

If Kylie sells 3 CDs (2 more)

$$W = \$0.75 \times 3 + \$50$$

$$= \$52.25$$

$\therefore$ Kylie earns \$1.50 more.

Answer B

51 i $A = (77 - 32) \div 1.8$

$$= 25$$

(1 mark)

ii $F = 2(C + 12)$ ✓✓

(2 marks)

52 $w = 2y^3 - 1$

$$\therefore 13 = 2y^3 - 1$$

$$2y^3 = 14$$

$$y^3 = 7$$

$$y = \sqrt[3]{7}$$

Answer C

53 $w = \frac{15y}{y + 12}$

$$\therefore w = \frac{15 \times 7}{7 + 12}$$

$$= 5.526 \dots$$

$$\div 5.53 \text{ to 2 decimal places}$$

Answer A